

CHE 321 – Mid-Semester Exam

(Open Notes, Open Books, No Electronics)

Time: 12:00 – 1:15 pm

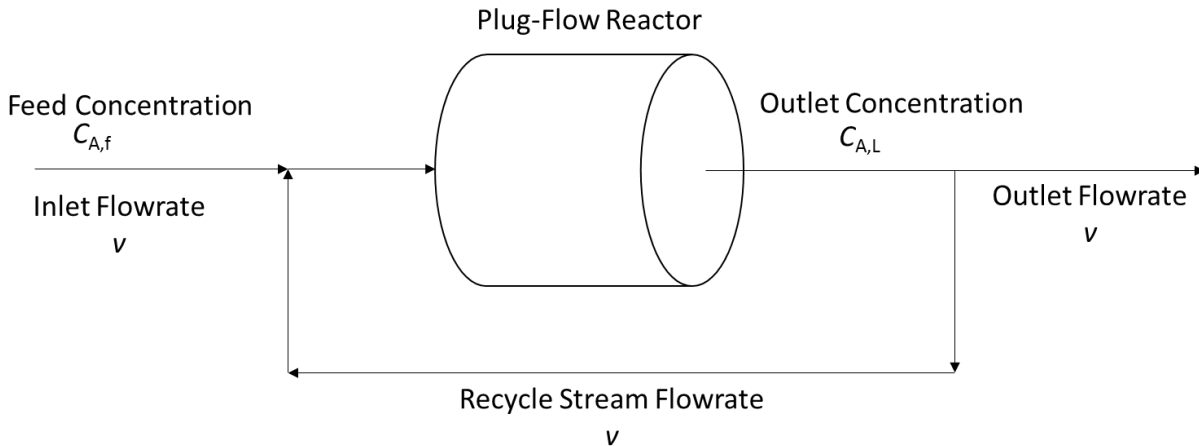
Name: _____

Question -1: (30 points) Circle one choice.

1. In a chemical reaction,
 - a. Total number of electrons are conserved
 - b. Total number of atoms are conserved
 - c. All of the above
2. An elementary reaction initially absorbs energy and then release energy to form products.
 - a. True
 - b. False
3. The order of elementary reaction is specific to
 - a. Reactants
 - b. Products
 - c. Both reactants and products
4. Transition state theory is applied to
 - a. Irreversible Reaction
 - b. Reversible Reaction
 - c. Elementary Reaction
5. Reaction invariant is a function of temperature.
 - a. True
 - b. False
6. For the non-isothermal reversible reaction, the change in moles of product with respect to time is always positive
 - a. True
 - b. False
7. The maximum value of reaction extent is determined by
 - a. Concentration of limiting reagent
 - b. Equilibrium constant
8. The reaction is autocatalytic when a product reacts with reactants to produce the same product.
 - a. True
 - b. False
9. Only first-order reactions can be lumped.
 - a. True
 - b. False
10. Effectiveness factor can increase with increasing Thiele modulus for non-isothermal reactions

- True
- False

Question-2: (35 points) Consider a plug-flow reactor with recycle as given in Figure below.

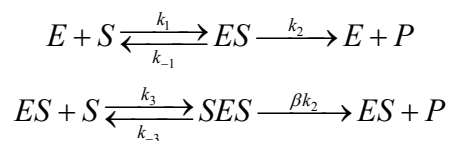


Schematic of Recycle Plug-Flow Reactor

A liquid phase reaction $2A \rightarrow B$ of 2nd order is conducted in a plug-flow reactor. The rate expression of the 2nd order reaction is given as $r = k C_A^2$. The feed contains only reactant A of concentration $C_{A,f}$ and it is mixed with recycle stream containing unconverted reactant of concentration $C_{A,L}$ before entering the plug-flow reactor. The flowrates of feed stream and recycle stream are identical and given as v . The plug-flow reactor has length L , cross-sectional area A , and volume V . The conversion of A in the recycle plug-flow reactor is 3/4.

- (15 points)** Write the steady-state continuity equation for A in the recycle plug-flow reactor. Integrate the continuity equation to show $\frac{kVC_{A,f}}{v} = \frac{8}{3}$.
- (20 points)** What would the conversion be if the recycle were shut off?

Question-3: (35 points) An enzyme may have more than one catalytically active center and the presence or absence of a substrate molecule at one site may influence the activity of the other sites. This is an example of allosteric substrate inhibition. Consider the simplest case of two identical active sites which is represented by the following reactions:



1. **(10 points)** Write the mole balance for the intermediate ES and SES and use the PSSH to obtain the expression for concentration of ES and SES .
2. **(5 points)** Write the enzyme balance equation for above reactions.
3. **(10 points)** Obtain the rate expression for the production of product P in terms of concentrations of enzyme and substrate.
4. **(10 points)** Qualitatively plot the reaction rate versus concentration of substrate for three cases- i) $\beta = 0$, ii) $\beta > 1$ and iii) $k_3 \ll k_{-3} + \beta k_2$